Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Check your working directory
- Load the tidyverse, lubridate, zoo, and trend packages
- Set your ggplot theme

```
library(tidyverse); library(lubridate); library(zoo); library(trend)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'zoo'
##
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```r
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
Liz_theme <- theme_grey (base_size = 10) +
  theme(
    plot.title = element_text(
      size = 14,
      face = "bold",
      hjust = 0.5,
      margin = margin(b=10), color = "#FAF0E6"),
    axis.title.x = element_text(color = "#FAF0E6", size = 10, face = "bold"),
    axis.title.y = element_text(color = "#FAF0E6", size = 10, face = "bold"),
    axis.text.x = element_text(color = "#FAF0E6"),
    axis.text.y = element_text(color = "#FAF0E6"),
    plot.background = element_rect(fill = "#008080"),
    panel.background = element_rect(fill = "#FAF0E6"),
    legend.position = "right",
    legend.title = element_text(
      face = "bold")
  )

theme_set(Liz_theme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named `GaringerOzone` of 3589 observation and 20 variables.

```r
#1
Ozone_TimeSeries.Raw <- dir(here('Data','Raw', 'Ozone_TimeSeries'),
pattern ="^EPAair_O3.*csv",
full.names = TRUE)
print(Ozone_TimeSeries.Raw)
```

```
##  [1] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2010_raw.csv"
##  [2] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2011_raw.csv"
##  [3] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2012_raw.csv"
##  [4] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2013_raw.csv"
##  [5] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2014_raw.csv"
##  [6] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2015_raw.csv"
##  [7] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2016_raw.csv"
##  [8] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2017_raw.csv"
##  [9] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2018_raw.csv"
## [10] "/home/guest/EDE_Fall2025/Data/Raw/Ozone_TimeSeries/EPAair_O3_GaringerNC2019_raw.csv"
```

```r
GaringerOzone <- Ozone_TimeSeries.Raw %>%
  map_dfr(read_csv)
dim(GaringerOzone)
```

```
## [1] 3589   20
```

```r
glimpse(GaringerOzone)
```

```
## Rows: 3,589
## Columns: 20
## $ Date                                 <chr> "01/01/2010", "01/02/2010", "01~
## $ Source                               <chr> "AQS", "AQS", "AQS", "AQS", "AQ~
## $ `Site ID`                            <dbl> 371190041, 371190041, 371190041~
## $ POC                                  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ `Daily Max 8-hour Ozone Concentration` <dbl> 0.031, 0.033, 0.035, 0.031, 0.0~
## $ UNITS                                <chr> "ppm", "ppm", "ppm", "ppm", "pp~
## $ DAILY_AQI_VALUE                      <dbl> 29, 31, 32, 29, 25, 31, 32, 30,~
## $ `Site Name`                          <chr> "Garinger High School", "Garing~
## $ DAILY_OBS_COUNT                      <dbl> 17, 17, 17, 17, 17, 17, 17, 17,~
## $ PERCENT_COMPLETE                     <dbl> 100, 100, 100, 100, 100, 100, 1~
## $ AQS_PARAMETER_CODE                   <dbl> 44201, 44201, 44201, 44201, 442~
## $ AQS_PARAMETER_DESC                   <chr> "Ozone", "Ozone", "Ozone", "Ozo~
## $ CBSA_CODE                            <dbl> 16740, 16740, 16740, 16740, 167~
## $ CBSA_NAME                            <chr> "Charlotte-Concord-Gastonia, NC~
## $ STATE_CODE                           <dbl> 37, 37, 37, 37, 37, 37, 37, 37,~
## $ STATE                                <chr> "North Carolina", "North Caroli~
## $ COUNTY_CODE                          <dbl> 119, 119, 119, 119, 119, 119, 1~
## $ COUNTY                               <chr> "Mecklenburg", "Mecklenburg", "~
## $ SITE_LATITUDE                        <dbl> 35.2401, 35.2401, 35.2401, 35.2~
## $ SITE_LONGITUDE                       <dbl> -80.78568, -80.78568, -80.78568~
```

Wrangle

3. Set your date column as a date class.

4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY__AQI__VALUE.

5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```r
# 3
GaringerOzone$Date <-mdy(GaringerOzone$Date)
class(GaringerOzone$Date)
```

```
## [1] "Date"
```

```r
# 4
GaringerOzone.Processed <- GaringerOzone %>%
  select(Date, `Daily Max 8-hour Ozone Concentration`, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(
  from = as.Date("2010-01-01"),
  to = as.Date("2019-12-31"),
  by = "day"
))

colnames(Days) <- "Date"

# 6
GaringerOzone <- Days %>%
  left_join(GaringerOzone.Processed, by = "Date")
dim(GaringerOzone)
```

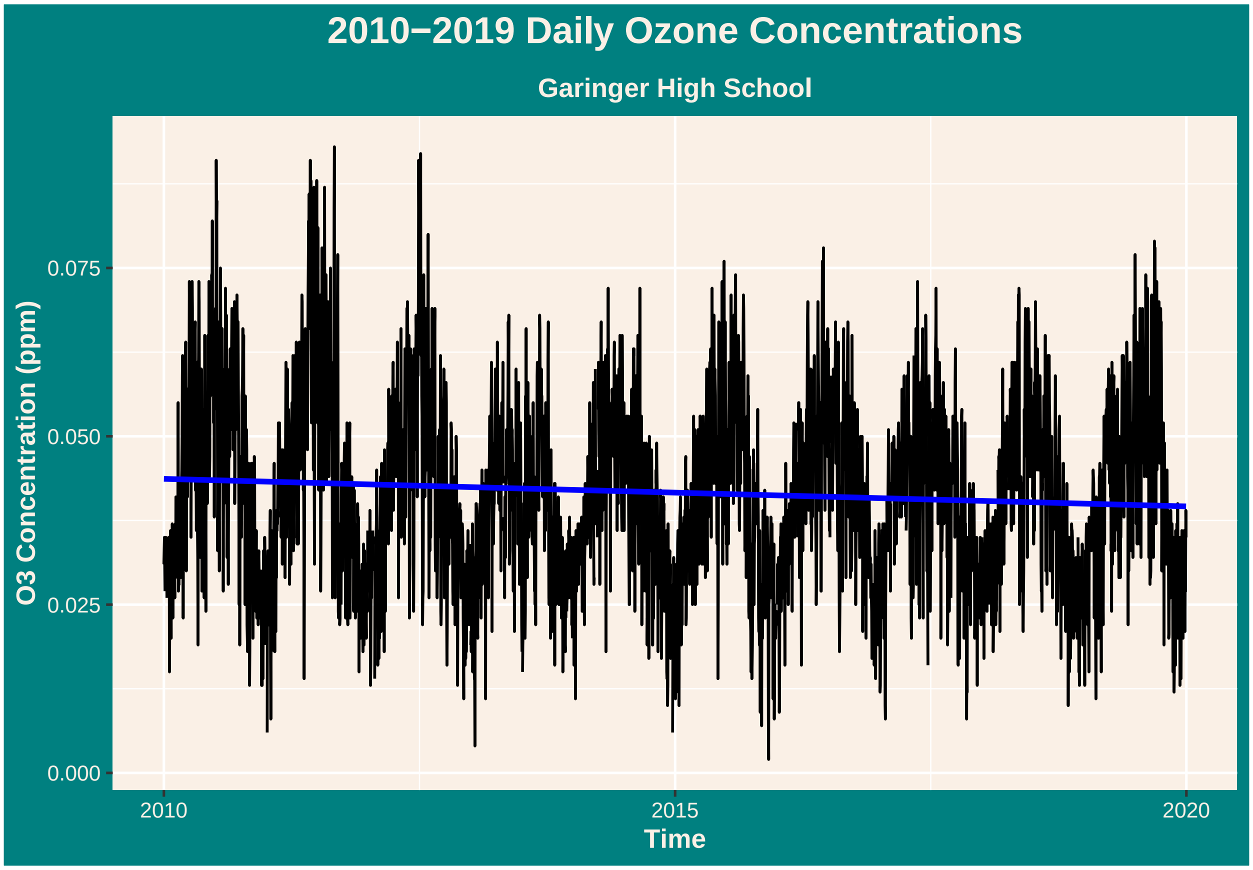
```
## [1] 3652    3
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```r
#7
O3.time.plot <- ggplot(GaringerOzone, aes(x = Date, y = `Daily Max 8-hour Ozone Concentration`)) +
  geom_line()+
  geom_smooth (method = "lm", se = FALSE, col = 'blue') +
  labs(
    title = "2010-2019 Daily Ozone Concentrations",
    subtitle = "Garinger High School",
    x= "Time",
    y = "O3 Concentration (ppm)"
  ) +
  theme (
    plot.subtitle = element_text(
      size = 10,
      face = "bold",
      hjust = 0.5,
      color = "#FAF0E6"))
print(O3.time.plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 63 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```



Answer: The linear trend line shows a slightly downward trend which suggests that ozone has a very gradual, slight decrease in concentrations over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <- GaringerOzone %>%
  arrange(Date) %>%
  mutate(Ozone.clean = zoo::na.approx(`Daily Max 8-hour Ozone Concentration`, x = Date, na.rm = FALSE))

summary(GaringerOzone$`Daily Max 8-hour Ozone Concentration`)
```

```
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
## 0.00200 0.03200 0.04100 0.04163 0.05100 0.09300      63
```

```
summary(GaringerOzone$Ozone.clean)
```

```
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

Answer: We didn't use piecewise constane because it assumes that missing values are equal to the last observed value nearest to that date (earlier or later). Spline interpolation would not have been the best method due to a quadratic function used to interpolate, adding unrealistic fluctuations in the data, rather than drawing a simple, realistic, straight line.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

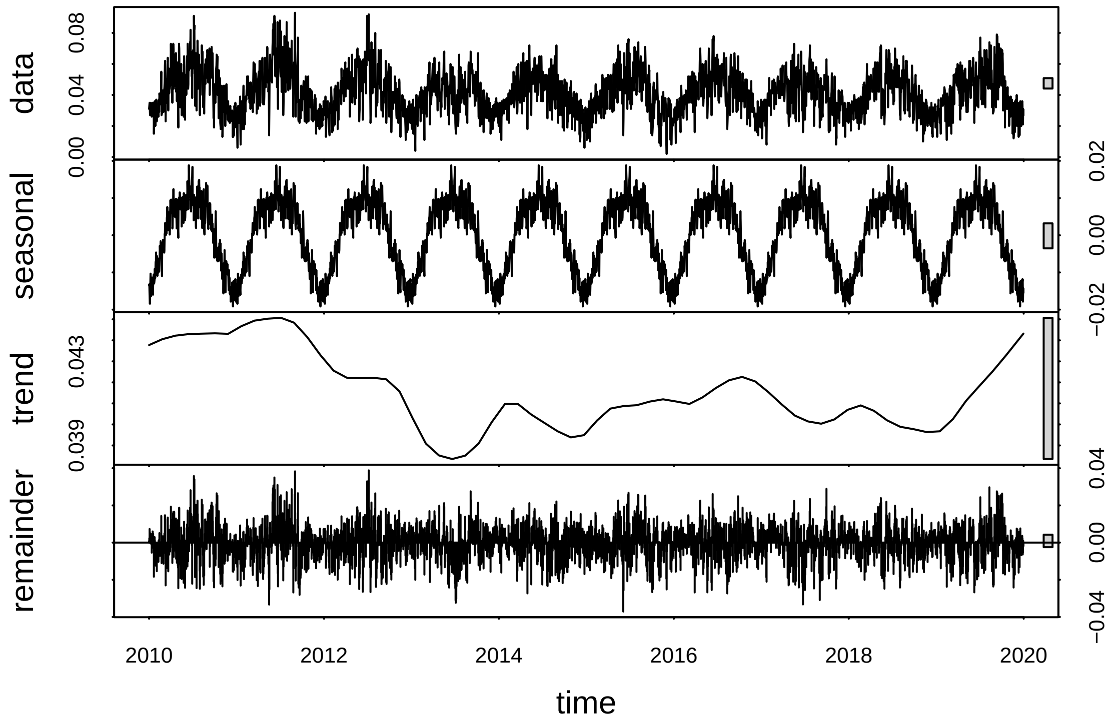
```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(year = year(Date),
    month = month(Date)) %>%
  group_by(year, month) %>%
  summarise(mean_ozone = mean(`Daily Max 8-hour Ozone Concentration`, na.rm = TRUE),
    .groups = "drop") %>%
  mutate(
    Date = as.Date(paste(year, month, "01", sep = "-")),
    mean_ozone.clean = zoo::na.approx(mean_ozone, x = Date, na.rm = FALSE))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

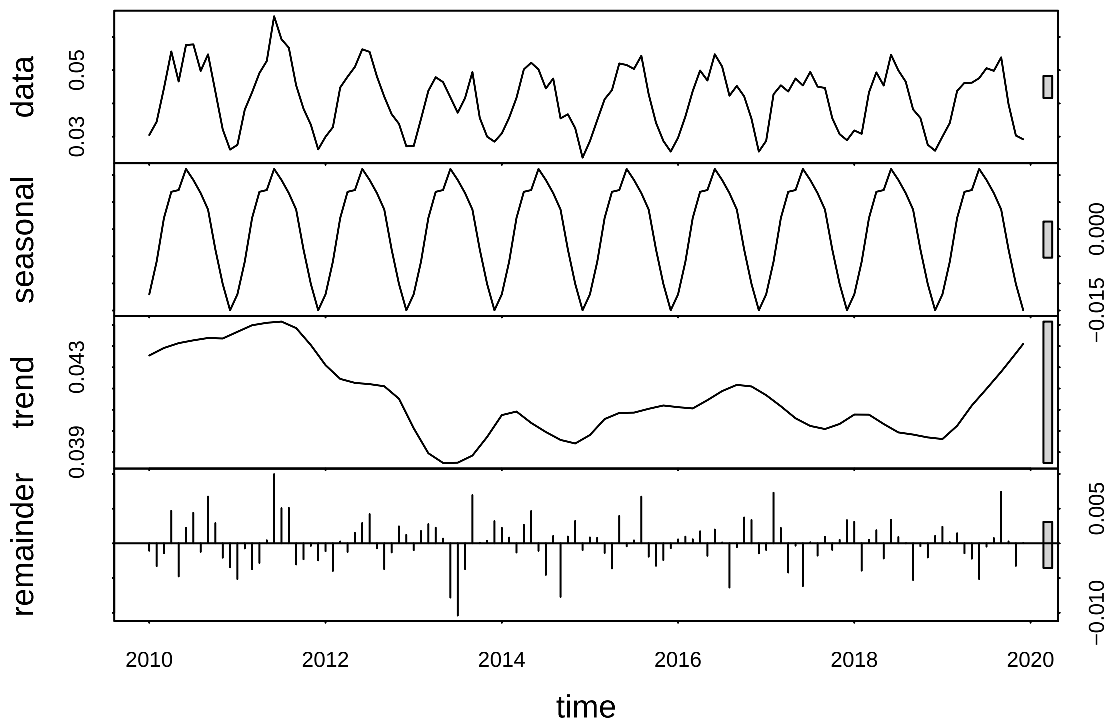
```
#10
start_date <- first(GaringerOzone$Date)
end_date   <- last(GaringerOzone$Date)

start_year <- year(start_date)
start_day  <- yday(start_date)

end_year <- year(end_date)
end_day  <- yday(end_date)

GaringerOzone.daily.ts <- ts(
  GaringerOzone$Ozone.clean,
  start = c(start_year, start_day),
  end   = c(end_year, end_day),
  frequency = 365)


f_month <- month(first(GaringerOzone.monthly$Date))
f_year  <- year(first(GaringerOzone.monthly$Date))

l_month <- month(last(GaringerOzone.monthly$Date))
l_year  <- year(last(GaringerOzone.monthly$Date))

GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$mean_ozone,
                               start = c(f_year, f_month),
                               end   = c(l_year, l_month),
                               frequency = 12)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```r
#11
GaringerOzone.daily.decompose <- stl(GaringerOzone.daily.ts,
                                     s.window = "periodic")
plot(GaringerOzone.daily.decompose)
```




```r
GaringerOzone.monthly.decompose <- stl(GaringerOzone.monthly.ts,
                                       s.window = "periodic")
plot(GaringerOzone.monthly.decompose)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
GaringerOzone.monthly.trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

GaringerOzone.monthly.trend
```

```
## tau = -0.163, 2-sided pvalue =0.022986
```

```
summary(GaringerOzone.monthly.trend)
```

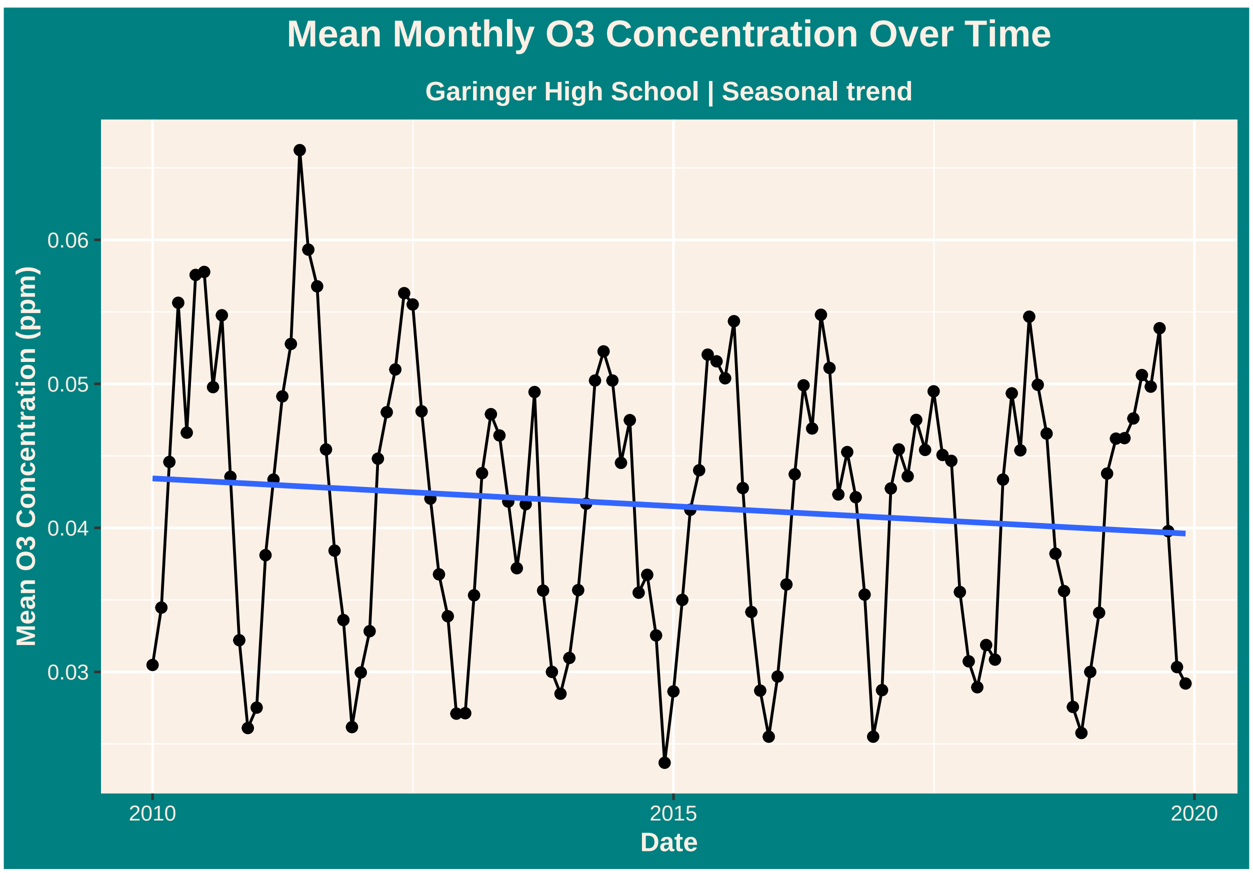
```
## Score =  -88 , Var(Score) = 1498
## denominator =  538.9944
## tau = -0.163, 2-sided pvalue =0.022986
```

Answer: Ozone concentrations has a seasonal cycle as the amount of solar energy inputs change (less during winter months, and more during summer months) throughout the year which helps convert gaseous oxygen molecule (O2) to create Ozone (O3). Season Mann-kendall test is ideal to test whether a monotonic trend exists if ozone concentrations has changed over the time including the seasonal component.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a geom_point and a geom_line layer. Edit your axis labels accordingly.

```r
# 13
O3.monthly.trend.plot <-ggplot(GaringerOzone.monthly,
                               aes(x = Date, y = mean_ozone.clean)) +
  geom_point() +
  geom_line() +
  labs (title = "Mean Monthly O3 Concentration Over Time",
        subtitle = "Garinger High School | Seasonal trend",
        y = "Mean O3 Concentration (ppm)") +
  geom_smooth (method = lm, se = FALSE) +
  theme (
    plot.subtitle = element_text(
      size = 10,
      face = "bold",
      hjust = 0.5,
      color = "#FAF0E6"))
print (O3.monthly.trend.plot)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

    Answer: The Mann-Kendall test results showed a tau value of -0.163 which indicates a downward negative trend in average ozone concentrations over time. This is supported by the blue trendline in the graph which shows a gradual, slight decline in mean concentrations. The p-value is 0.022986 which proves that our data shows 95% statistical signifigance. Furthermore, our understanding that average ozone concentrations is a seasonal variable is supported in the graph as the points connect in relatively repeating fluxuating curves across each annual period (peaks towards the middle months of each year and dips towards the start/end months of each year).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the EnoDischarge on the lesson Rmd file.

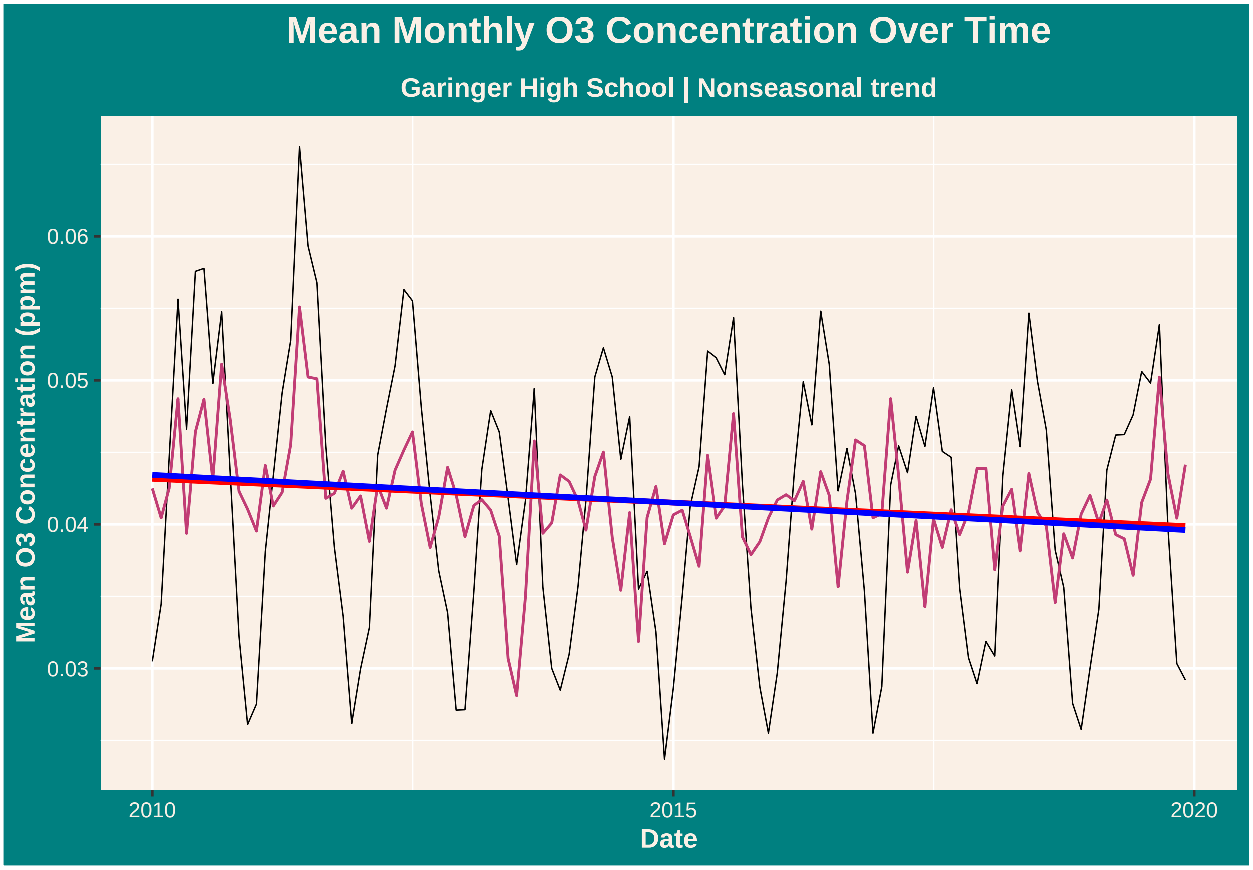
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.month.nonseasonal <- GaringerOzone.monthly.ts - GaringerOzone.monthly.decompose$time.seri

GaringerOzone.monthly.new <- GaringerOzone.monthly %>%
  mutate(Nonseasonal = as.numeric(GaringerOzone.month.nonseasonal))

nonseasonal.plot <- ggplot(GaringerOzone.monthly.new, aes(x = Date)) +
  geom_line(aes(y = mean_ozone), size = 0.25) +
  geom_line(aes(y = Nonseasonal), color = "#c13d75ff") +
  labs(title = "Mean Monthly O3 Concentration Over Time",
       subtitle = "Garinger High School | Nonseasonal trend",
       y = "Mean O3 Concentration (ppm)") +
  geom_smooth(aes(y = Nonseasonal), method = lm, se = FALSE, color = "red") +
  geom_smooth(aes(y = mean_ozone), method = lm, se = FALSE, color = "blue") +
  theme(
    plot.subtitle = element_text(
      size = 10,
      face = "bold",
      hjust = 0.5,
      color = "#FAF0E6"))
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
print(nonseasonal.plot)
```

```
## ‘geom_smooth()‘ using formula = ’y ~ x’
## ‘geom_smooth()‘ using formula = ’y ~ x’
```




```r
#16
nonseasonal.trend <- trend::mk.test(GaringerOzone.month.nonseasonal)
nonseasonal.trend
```

```
##
##  Mann-Kendall trend test
##
## data:  GaringerOzone.month.nonseasonal
## z = -2.8966, n = 120, p-value = 0.003773
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##             S          varS           tau
## -1.278000e+03  1.943647e+05 -1.790167e-01
```

> Answer: The Mann-Kenndall tau value changed from -0.163 (seasonal) to -0.179 (nonseasonal) which showcases a stronger negative correlation (slighter greater decline) between ozone concentration over time. We found that the nonseasonal trend has higher statistical signifigance (0.0038<0.05) than the seasonal p-value of 0.0230 - although both models show 95% statistical confidence overall and a negative trend in ozone concentrations over time.